

Shiyue(Jason) Hou

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Summary

Ph.D. candidate in Computer Engineering focused on agent-based architecture, federated learning frameworks, and distributed data infrastructure. Experienced in designing end-to-end ML systems for real-world workloads, including intelligent storage systems, adaptive caching and prefetching, and system-aware learning frameworks.

Education

- 2021-present **PhD, Computer Engineering, Northeastern University, Boston, MA.**
AI Agent Systems, Federated Learning for system, Data-intensive Systems, Intelligent Storage Systems, Caching and Prefetching, I/O Workload Modeling, System-aware Machine Learning
- 2015-2018 **Master of Engineering, Electrical Engineering, Lamar University, Beaumont, TX.**
- 2011-2015 **Bachelor of Engineering, Electrical Engineering, Inner Mongolia University, China.**

Work Experience

- June 2024 - **Research Associate Intern, Hewlett Packard Labs, Milpitas, CA, United States.**
Present
 - Designed an AI-agent-based framework for adaptive data placement in distributed storage systems, combining RL-based importance prediction, cost-aware admission control, and feedback-driven optimization across heterogeneous storage tiers.
 - Designed a graph neural network-based storage prefetching framework that models temporal and structural I/O access patterns, enabling scalable federated training across heterogeneous storage nodes and significantly improving cache efficiency.
 - Led the design of an FL-LSTM storage optimization framework that combines federated learning and time-series forecasting to drive intelligent cache management, improving prediction quality while reducing energy and carbon costs at scale.
- May 2023 - **Visiting Researcher, Pacific Northwest National Laboratory, Richland, WA, United States.**
September 2025
 - Built a near-data processing framework on Ceph storage systems to offload data analytics and machine learning operations to storage nodes, reducing data movement and alleviating compute bottlenecks for large-scale AI workloads.
 - Developed data movement-aware ML pipelines using Apache Arrow, Parquet, and DuckDB, improving end-to-end throughput, resource utilization, and scalability across distributed storage and computing infrastructures.
- June 2022 - **System Performance Intern, NetApp, Inc, San Jose, CA, United States.**
September 2022
 - Re-architected a legacy performance testing framework in Python with integrated ML-based anomaly detection, enabling scalable and data-driven performance analysis.
 - Built automated testing and telemetry pipelines for ONTAP replication systems, supporting large-scale data collection and downstream ML-driven optimization.
- December 2020 - July 2021 **Senior Software Engineer, Bilibili Group, Shanghai, China.**
 - Developed cross-platform UWP applications for Windows, Surface, and Xbox, supporting large-scale user-facing systems.
 - Built data collection and telemetry mechanisms in C#, enabling user behavior tracking and supporting downstream data analysis workflows.
 - Designed cross-platform data integration pipelines aligned with Android and iOS systems for consistent analytics and debugging.

August 2018 - **Software Engineer**, Intel Corporation, Dalian, China.

- October 2020
- Developed data conversion and visualization tools for 3D NAND manufacturing systems using C#, SQL, and PL/SQL, enabling efficient integration of multi-source data.
 - Built ASP.NET-based data analytics services for factory performance monitoring and forecasting, supporting large-scale manufacturing operations.
 - Designed multi-threaded data pipelines and asynchronous communication mechanisms to improve system performance and reduce operational errors.

Publications

- 2025 **Shiyue Hou**, Pavana Prakash, Rolando Pablo Hong Enriquez, Ningfang Mi, Alex Veprinsky, Dejan Milojicic, Carbon Friendly Federated Learning Framework for Optimized Distributed Storage Systems, In *ACM SIGMETRICS Performance Evaluation Review*.
- 2025 Allen Yang, **Shiyue Hou**, Yiming Xie, Bo Sheng, Ningfang Mi, Probabilistic Evaluation-based Client Selection for Federated Learning with Overlapping Clients, In *IEEE International Conference on High Performance Computing and Communications (HPCC)*.
- 2024 **Shiyue Hou**, Zhenglun Kong, FedLDCS: Adaptive Divergence-Based Client Selection for Federated Learning, In *International Joint Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD)*.
- 2024 **Shiyue Hou**, Nathan R. Tallent, Li Wang, Ningfang Mi, Performance Analysis of Data Processing in Distributed File Systems with Near Data Processing, In *International Symposium on Networks, Computers and Communications (ISNCC)*.
- 2024 Li Wang, **Shiyue Hou**, Yiming Xie, Ningfang Mi, Uncovering the Impact of Bursty Workloads on System Performance in Serverless Computing, In *International Symposium on Networks, Computers and Communications (ISNCC)*.
- 2018 Yao, Qingjiang, Praphul Joshi, Chiung-fang Chang, Chelsea McDonalds, Jason Tran, William Wheeler, **Shiyue Hou**, Advocating a New Approach to Governing Water, Energy, and Food Security, In *Journal of International Crisis and Risk Communication Research*.

Preprints

- 2026 **Shiyue Hou**, Pavana Prakash, Rolando Pablo Hong Enriquez, Alex Veprinsky, Dejan Milojicic, Ningfang Mi, TierPilot: A Unified Learning Framework for Data Placement in Distributed Storage Systems, Under Review.
- 2026 **Shiyue Hou**, Pavana Prakash, Rolando Pablo Hong Enriquez, Alex Veprinsky, Dejan Milojicic, Ningfang Mi, FedEcoStream: Energy-Aware Federated Learning with Stream-Based Prefetching for Distributed Storage Optimization, Under Review.
- 2024 Zhenglun Kong, Zheng Zhan, **Shiyue Hou**, Yifan Gong, Xin Meng, Peiyan Dong, Xuan Shen, Zifeng Wang, Pu Zhao, Hao Tang, Stratis Ioannidis, Yanzhi Wang, Enabling Flexible Multi-LLM Integration for Scalable Knowledge Aggregation, Under Review.
- 2024 Jiaqi Wu, Simin Chen, Yuzhe Yang, Yijiang Li, **Shiyue Hou**, Rui Jing, Zehua Wang, Wei Chen, Zijian Tian, FedDTPT: Federated Discrete and Transferable Prompt Tuning for Black-Box Large Language Models, Under Review.

Patents

- 2026 **Shiyue Hou**, Pavana Prakash, Rolando Pablo Hong Enriquez, Alex Veprinsky, Dejan Milojicic, Adaptive Delta LBA Graph Rewriting for Storage Prefetching, Filing.
- 2026 **Shiyue Hou**, Pavana Prakash, Rolando Pablo Hong Enriquez, Alex Veprinsky, Dejan Milojicic, Divergence-aware Client Selection in Hierarchical Federated Learning for Distributed Storage Systems, Filing.

Professional Activities

- Peer Review
 - Conference: NeurIPS 2026, 2024, AAAI 2027, 2026, ICLR 2026, MIDL 2025
 - Journal: IEEE TNNLS 2024, IEEE Access 2024, 2023, IEEE Big Data 2023
- Invited Talks
 - A Global In-Memory Cache and Computation Tier for DAOS, Parallel Data Systems Workshop, SC 2024, Atlanta, Nov 2024
 - Developing a Python Module for Accessing a Global Distributed Cache, Hewlett Packard Enterprise, San Jose, CA, Aug 2024
 - Understanding Emerging Storage Systems for Near-Data Processing with Machine Learning, Pacific Northwest National Laboratory, Richland, WA, Aug 2023
- Service Service for SC'27, SC'26, HotStorage'23
- Leadership Session Chair, International Conference on High Performance Computing and Communications (HPCC), Exeter, UK, 2025

Academic Achievements & Recognitions

- 2025 SIGMETRICS Conference Student Travel Grant Award
- 2021 Research Assistant Fellowship, College of Engineering, Northeastern University

Computer skills

- ML Systems PyTorch, TensorFlow, Ray, PyG, Scikit-learn
- Distributed Systems Ceph, Spark, libvirt, CXL, KVM, QEMU, Docker
- Data Systems PyArrow, DuckDB, Petastorm, Pandas, SQL,
- Programming Python, C++, C#, Bash, Java

Work Authorization

Eligible to work in the U.S. and do not require future sponsorship.